Basis and Dimension Reading

Now that we have developed the concepts of span and linear independence in the context of a vector space, we are ready to consider the notion of basis and dimension in the same setting. As with much of this chapter, the definitions of basis and dimension are essentially the same as in Euclidean space. Let's start with basis.

DEFINITION 10

Let $\mathcal{V}$ be a subset of a vector space V. Then $\mathcal{V}$ is a basis of V if $\mathcal{V}$ is linearly independent and spans V.

Example 1

Is the set $\left\{x^2+4x-3,\ x^2+1,\ x-2\right\}$ a basis for $\mathbf{P}^2$?

Solution

We need to determine if the given set is linearly independent and spans $\mathbf{P}^2$. Of the two, span is generally more difficult to verify, so let's tackle that first. (We will also find out about linear independence along the way.) A typical vector in $\mathbf{P}^2$ has the form $a_2x^2+a_1x+a_0$, and for each such vector we need to know if there exist corresponding scalars c_1, c_2, and c_3 such that

$$c_1\left(x^2+4x-3\right)+c_2\left(x^2+1\right)+c_3(x-2)=a_2x^2+a_1x+a_0$$

Reorganizing the left side to collect common terms, we have

$$\left(c_1+c_2\right)x^2+\left(4c_1+c_3\right)x+\left(-3c_1+c_2-2c_3\right)=a_2x^2+a_1x+a_0$$

Comparing coefficients gives us the linear system

$$\begin{array}{rcrcrcl} c_1 & + & c_2 & & & = & a_2 \\ 4c_1 & & & + & c_3 & = & a_1 \\ -3c_1 & + & c_2 & - & 2c_3 & = & a_0 \end{array} \qquad (1)$$

Using our standard solution methods, we find that for each choice of a_0, a_1, and a_2, the system has unique solution

$c_1=\frac{-a_0+2a_1+a_2}{6}$, $c_2=\frac{5a_0-2a_1-a_2}{6}$, $c_3=a_2-a_1-a_0$

Applying both parts of Theorem 9 in Lesson 2, Span and Linear Independence, we can conclude that our set is both linearly independent and spans $\mathbf{P}^2$. Hence the set is a basis of $\mathbf{P}^2$.

The method of solution in Example 1 suggests the following general theorem.

THEOREM 11

The set $\mathcal{V}=\left\{\mathbf{v}_1, \ldots, \mathbf{v}_m\right\}$ is a basis for a vector space V if and only if the equation

$c_1\mathbf{v}_1+\ldots+c_m\mathbf{v}_m=\mathbf{v}$ (2)

has a unique solution $c_1, \ldots, c_m$ for every $\mathbf{v}$ in V.

The proof follows from applying Theorem 9 in Lesson 2, and is left as an exercise.

Example 2

Verify that the set $\left\{1,\ x,\ x^2, \ldots, x^n\right\}$ is a basis for $\mathbf{P}^n$. (This is called the **standard basis** for $\mathbf{P}^n$.)

Solution

Suppose that $a_nx^n+\ldots+a_1x+a_0$ is a typical vector in $\mathbf{P}^n$. Forming a linear combination of our set and setting it equal to our typical vector produces

$c_1(1)+c_2(x)+\ldots+c_{n+1}\left(x^n\right)=a_nx^n+\ldots+a_1x+a_0$

Comparing coefficients instantly shows that the unique linear combination is given by $c_1=a_0, c_2=a_1, \ldots, c_{n+1}=a_n$. Since this works for every vector in $\mathbf{P}^n$, by Theorem 11 our set forms a basis for $\mathbf{P}^n$.

Setting $n=2$ in Example 2 gives another basis for $\mathbf{P}^2$. Note that this basis has the same number of elements as the basis in Example 1, illustrating the following theorem, which carries over from Euclidean space.

THEOREM 12

Suppose that $\mathcal{V}_1$ and $\mathcal{V}_2$ are both bases of a vector space V. Then $\mathcal{V}_1$ and $\mathcal{V}_2$ have the same number of elements.

The proof is left as an exercise. The theorem also applies to vector spaces with bases that have infinitely many vectors: if one basis has infinitely many vectors, then they all do.

Example 3

Show that every basis for $\mathbf{P}$ has infinitely many elements.

Solution

In Example 7 of Lesson 2, it is shown that the set $\left\{1,\ x,\ x^2, \ldots\right\}$ is linearly independent, and in a later exercise, it is shown that $\left\{1,\ x,\ x^2, \ldots\right\}$ spans $\mathbf{P}$. Therefore the set $\left\{1,\ x,\ x^2, \ldots\right\}$ is a basis for $\mathbf{P}$, and thus by Theorem 12 every basis for $\mathbf{P}$ must have infinitely many elements.

Since every basis for a vector space has the same number of vectors, the following definition makes sense.

DEFINITION 13

The dimension of a vector space V is equal to the number of vectors in any basis of V. If a basis of V has infinitely many vectors, then we say that the dimension is infinite.

For example, we have seen that $\mathbf{P}^n$ has dimension $n+1$ and $\mathbf{P}$ has infinite dimension. Put briefly, $\dim\left(\mathbf{P}^n\right)=n+1$ and $\dim(\mathbf{P})=\infty$.

Note that a trivial vector space $V=\{\mathbf{0}\}$ consisting only of the zero vector has no basis because it has no linearly independent subsets. We define $\dim(V)=0$ when V is a trivial vector space.

Example 4

Find the dimension of $\mathbf{R}^{2\times2}$, the vector space of real 2×2 matrices.

Solution

All we need to do is find any basis for $\mathbf{R}^{2\times2}$ and then count the vectors. The standard basis is given by

$$\left\{\begin{bmatrix}1&0\\0&0\end{bmatrix},\begin{bmatrix}0&1\\0&0\end{bmatrix},\begin{bmatrix}0&0\\1&0\end{bmatrix},\begin{bmatrix}0&0\\0&1\end{bmatrix}\right\}$$

Verification that this is a basis for $\mathbf{R}^{2\times2}$ is left as an exercise. Hence it follows that $\dim\left(\mathbf{R}^{2\times2}\right)=4$.

A similar argument to the one in Example 4 can be used to show that $\dim\left(\mathbf{R}^{n\times m}\right)=nm$.

Since subspaces are essentially vector spaces within vector spaces, they also have dimensions.

Example 5

Find the dimension of the subspace $S=\operatorname{span}\{T_1,\ T_2,\ T_3\}$ of $T(2,\ 2)$, where

$$T_1\left(\begin{bmatrix}x_1\\x_2\end{bmatrix}\right)=\begin{bmatrix}x_1-x_2\\2x_1\end{bmatrix},\ T_2\left(\begin{bmatrix}x_1\\x_2\end{bmatrix}\right)=\begin{bmatrix}x_2\\2x_1+x_2\end{bmatrix},\ T_3\left(\begin{bmatrix}x_1\\x_2\end{bmatrix}\right)=\begin{bmatrix}2x_1-5x_2\\-2x_1-3x_2\end{bmatrix}$$

Solution

Let's check if the set is linearly independent. If so, then the dimension is 3 and we are done. If not, then we will identify a linear relation among the vectors that will allow us to remove one.

The zero vector in $T(2,\ 2)$ is the linear transformation $T_0(\mathbf{x})=\mathbf{0}$. We need to find the values of c_1, c_2, and c_3 such that

$c_1T_1(\mathbf{x})+c_2T_2(\mathbf{x})+c_3T_3(\mathbf{x})=T_0(\mathbf{x})$

is true for all $\mathbf{x}$ in $\mathbf{R}^2$. This is equivalent to the equation

$$c_1\begin{bmatrix}x_1-x_2\\2x_1\end{bmatrix}+c_2\begin{bmatrix}x_2\\2x_1+x_2\end{bmatrix}+c_3\begin{bmatrix}2x_1-5x_2\\-2x_1-3x_2\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$$

which in turn is equivalent to the system

$$\begin{array}{l} c_1\left(x_1-x_2\right)+c_2x_2+c_3\left(2x_1-5x_2\right)=0 \\ c_1\left(2x_1\right)+c_2\left(2x_1+x_2\right)+c_3\left(-2x_1-3x_2\right)=0 \end{array}$$

Reorganizing to separate out x_1 and x_2 gives us

$$\begin{array}{l} x_1\left(c_1+2c_3\right)+x_2\left(-c_1+c_2-5c_3\right)=0 \\ x_1\left(2c_1+2c_2-2c_3\right)+x_2\left(c_2-3c_3\right)=0 \end{array}$$

Since the system must be satisfied for all values of x_1 and x_2, we require that

$$\begin{array}{rcrcrcl} c_1 & & & + & 2c_3 & = & 0 \\ -c_1 & + & c_2 & - & 5c_3 & = & 0 \\ 2c_1 & + & 2c_2 & - & 2c_3 & = & 0 \\ & & c_2 & - & 3c_3 & = & 0 \end{array}$$

Using our standard methods, we can show that this system has infinitely many solutions, among them $c_1=2$, $c_2=-3$, and $c_3=-1$. Therefore

$2T_1(\mathbf{x})-3T_2(\mathbf{x})-T_3(\mathbf{x})=T_0(\mathbf{x})$

or

$2T_1(\mathbf{x})-3T_2(\mathbf{x})=T_3(\mathbf{x})$

Since T_3 is a linear combination of T_1 and T_2, it follows that $S=\operatorname{span}\{T_1,\ T_2\}$. It is straightforward to verify that T_1 and T_2 are not multiples of each other and so are linearly independent. Hence $\{T_1,\ T_2\}$ is a basis for S, and therefore $\dim(S)=2$.

In Example 5, we started with a set that spanned a subspace S and removed a vector to form a basis for S. The following theorem formalizes this process, along with the process of expanding a linearly independent set to a basis.

THEOREM 14

Let $\mathcal{V}=\{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$ be a subset of a nontrivial finite dimensional vector space V.

(a) If $\mathcal{V}$ spans V, then either $\mathcal{V}$ is a basis for V or vectors can be removed from $\mathcal{V}$ to form a basis for V.

(b) If $\mathcal{V}$ is linearly independent, then either $\mathcal{V}$ is a basis for V or vectors can be added to $\mathcal{V}$ to form a basis for V.

The proof is left as an exercise. Note that Theorem 14 also applies to subspaces of vector spaces.

Example 6

Extend the set $\mathcal{V}=\left\{x^2+2x+1,\ x^2+6x+3\right\}$ to a basis for $\mathbf{P}^2$.

Solution

We solve this by starting with a set we know spans $\mathbf{P}^2$ and then applying Theorem 14(b). We know that $\left\{x^2,\ x,\ 1\right\}$ is a basis for $\mathbf{P}^2$, so the combined set

$\left\{x^2+2x+1,\ x^2+6x+3,\ x^2,\ x,\ 1\right\}$ (3)

spans $\mathbf{P}^2$. To find a basis that includes $\mathcal{V}$, we need to determine the dependences among the vectors in (3). We start with the equation

$c_1\left(x^2+2x+1\right)+c_2\left(x^2+6x+3\right)+c_3x^2+c_4x+c_5=0$

which is equivalent to

$\left(c_1+c_2+c_3\right)x^2+\left(2c_1+6c_2+c_4\right)x+\left(c_1+3c_2+c_5\right)=0$

Setting each coefficient equal to zero yields the vector equation

$$c_1\begin{bmatrix}1\\2\\1\end{bmatrix}+c_2\begin{bmatrix}1\\6\\3\end{bmatrix}+c_3\begin{bmatrix}1\\0\\0\end{bmatrix}+c_4\begin{bmatrix}0\\1\\0\end{bmatrix}+c_5\begin{bmatrix}0\\0\\1\end{bmatrix}+=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

The augmented matrix and corresponding echelon form are

$$\left[\begin{array}{ccccc|c}1&1&1&0&0&0\\2&6&0&1&0&0\\1&3&0&0&1&0\end{array}\right]\sim\left[\begin{array}{ccccc|c}1&1&1&0&0&0\\0&2&0&-1&0&0\\0&0&0&0&1&0\end{array}\right]$$

Since the leading terms appear in columns 1, 2, and 4 of the echelon matrix, it follows that the vectors associated with c_1, c_2, and c_4 are linearly independent. To see why, note that if the vectors x^2 and 1 were eliminated from (3), then we would have the linear system

$$\begin{array}{rcrcrcl} c_1 & + & c_2 & & & = & 0 \\ 2c_1 & + & 6c_2 & + & c_4 & = & 0 \\ c_1 & + & 3c_2 & & & = & 0 \end{array}$$

The augmented matrix and echelon form are as before, but with the third and fifth columns removed. (The row operations are the same as before.)

$$\left[\begin{array}{ccc|c}1&1&0&0\\2&6&1&0\\1&3&0&0\end{array}\right]\sim\left[\begin{array}{ccc|c}1&1&0&0\\0&2&0&0\\0&0&1&0\end{array}\right]$$

We see that the new system has a unique solution, so the set

$\mathcal{V}_1=\left\{x^2+2x+1,\ x^2+6x+3,\ x\right\}$

is linearly independent. Furthermore, if $\mathcal{V}_1$ is not a basis, then by Theorem 14(b) we can add vectors to $\mathcal{V}_1$ to form a basis for $\mathbf{P}^2$. But this would mean that $\dim\left(\mathbf{P}^2\right)>3$, which we know is false. Therefore $\mathcal{V}_1$ is a basis.

Dimension gives us a way to measure the size of a vector space. Thus if one vector space is contained in another, it seems reasonable that the dimension of the former would be smaller than that of latter. This was true in Euclidean space, and is also true here.

THEOREM 15

Let V_1 and V_2 be vector spaces with V_1 a subset of V_2, and suppose both have the same definition of addition and scalar multiplication. Then $\dim(V_1)\leq\dim(V_2)$.

Proof

If $\dim(V_2)=\infty$, then the theorem is true regardless of the dimension of V_1. (Similar reasoning applies if $\dim(V_1)=0$.) Now suppose that $\dim(V_2)$ is finite, and let $\mathcal{V}_1$ be a basis for V_1. Since V_1 is a subset of V_2, $\mathcal{V}_1$ is a linearly independent subset of V_2. Thus, by Theorem 14(b), $\mathcal{V}_1$ is a basis for V_2 or can be expanded to a basis for V_2. Therefore the number of vectors in $\mathcal{V}_1$ is less than or equal to $\dim(V_2)$. Since $\mathcal{V}_1$ is a basis of V_1, we conclude that $\dim(V_1)\leq\dim(V_2)$.

Note that in Theorem 15, V_1 can also be considered a subspace of V_2.

Example 7

Show that $\dim(C(\mathbf{R}))=\infty$ without finding a basis for $C(\mathbf{R})$.

Solution

We have already shown that $\dim(\mathbf{P})=\infty$. Since all polynomials are continuous functions, then $\mathbf{P}$ is a subspace of $C(\mathbf{R})$. Hence by Theorem 15 we have $\dim(C(\mathbf{R}))=\infty$.

THEOREM 16

Let $\mathcal{V}=\{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$ be a subset of a vector space V with $\dim(V)=n$.

(a) If $m<n$, then $\mathcal{V}$ does not span V.

(b) If $m>n$, then $\mathcal{V}$ is linearly dependent.

The proof is left as an exercise. As with previous theorems in this lesson, Theorem 16 also applies to subspaces.

Example 8

Determine by inspection which of the subsets cannot span $\mathbf{R}^{2\times2}$ and which must be linearly dependent.

$$\begin{aligned}\mathcal{V}_1&=\left\{\begin{bmatrix}2&1\\4&3\end{bmatrix},\begin{bmatrix}5&0\\1&2\end{bmatrix},\begin{bmatrix}8&3\\5&6\end{bmatrix}\right\}\\ \mathcal{V}_2&=\left\{\begin{bmatrix}5&9\\0&4\end{bmatrix},\begin{bmatrix}7&6\\1&2\end{bmatrix},\begin{bmatrix}1&3\\7&8\end{bmatrix},\begin{bmatrix}6&2\\7&1\end{bmatrix}\right\}\\ \mathcal{V}_3&=\left\{\begin{bmatrix}3&6\\9&0\end{bmatrix},\begin{bmatrix}1&4\\3&8\end{bmatrix},\begin{bmatrix}3&2\\0&1\end{bmatrix},\begin{bmatrix}0&1\\6&6\end{bmatrix},\begin{bmatrix}3&2\\5&5\end{bmatrix}\right\}\end{aligned}$$

Solution

In Example 4, we showed that $\dim\left(\mathbf{R}^{2\times2}\right)=4$. Hence by Theorem 16(a), $\mathcal{V}_1$ cannot span $\mathbf{R}^{2\times2}$ because it has fewer than four vectors. By Theorem 16(b), $\mathcal{V}_3$ must be linearly dependent because it has more than four vectors.

Note that Theorem 16 cannot tell us if a set spans a vector space or is linearly independent. Hence we cannot conclude that $\mathcal{V}_2$ or $\mathcal{V}_3$ spans $\mathbf{R}^{2\times2}$, nor can we conclude that $\mathcal{V}_1$ or $\mathcal{V}_2$ is linearly independent.

THEOREM 17

Let $\mathcal{V}=\{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$ be a subset of a vector space V with $\dim(V)=m<\infty$. If $\mathcal{V}$ is linearly independent or spans V, then $\mathcal{V}$ is a basis for V.

Theorem 17 also applies to subspaces. This theorem tells us that if a set has the same number of vectors as the dimension of the vector space, then we need only verify either span or linear independence to determine if the set is a basis.

Example 9

Show that the set

$\mathcal{V}=\left\{x^3-x^2-5x+1,\ 2x^2+3x,\ x^3+3x^2=4x+2,\ -3x^2+5\right\}$

is a basis for $\mathbf{P}^3$.

Solution

We have $\dim\left(\mathbf{P}^3\right)=4$ and $\mathcal{V}$ has four elements, so by Theorem 17 we need only show $\mathcal{V}$ is linearly independent or spans $\mathbf{P}^3$ to prove that $\mathcal{V}$ is a basis for $\mathbf{P}^3$. Let's show that $\mathcal{V}$ is linearly independent. Suppose that c_1, c_2, c_3, and c_4 are scalars such that

$c_1\left(x^3-x^2-5x+1\right)+c_2\left(2x^2+3x\right)+c_3\left(x^3+3x^2-4x+2\right)+c_4\left(-3x^2+5\right)=0$

Reorganizing to collect common terms and setting each coefficient equal to zero gives us

$$\begin{array}{rcrcrcl} c_1 & & & + & c_3 & = & 0 \\ -c_1 & + & 2c_2 & + & 3c_3 & - & 3c_4 & = & 0 \\ -5c_1 & + & 3c_2 & - & 4c_3 & & & = & 0 \\ c_1 & & & + & 2c_3 & + & 5c_4 & = & 0 \end{array}$$

To determine linear independence, all we need to know is if this system has any nontrivial solutions. Instead of transforming to an augmented matrix and performing row operations, let's define the coefficient matrix

$$C=\begin{bmatrix}1&0&1&0\\-1&2&3&-3\\-5&3&-4&0\\1&0&2&5\end{bmatrix}$$

By applying techniques from earlier lessons, we can show that $\det(C)=-59$. Since the determinant of the coefficient matrix C is nonzero, our linear system has only the trivial solution. Therefore $\mathcal{V}$ is a linearly independent set and is a basis for $\mathbf{P}^3$.

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